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### UNDERCABINET LIGHTING DEVICES AND METHODS FOR INSTALLING THE SAME

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#### Abstract

A lighting device includes a housing, a light source, a light guide panel, and a lens. The housing includes a base, a first side wall, a second side wall, a first bumper protruding from an outer surface of the first side wall, and a second bumper protruding from an outer surface of the second side wall. The first bumper and second bumper to aid in positioning the housing within a slot formed in an external structure. The light guide panel is configured to diffuse light emitted by the light source. The lens is coupled to the housing and includes a base, a first side wall, a second side wall, a first wing protruding from an inner surface of the first side wall, and a second wing protruding from an inner surface of the second side wall.

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## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of, and priority to, U.S. Provisional Application No. 63/302,901, filed Jan. 25, 2022, which is hereby incorporated by reference herein in its entirety.

### **TECHNICAL FIELD**

[0002] The present disclosure relates generally to lighting devices, and more particularly, to lighting devices configured to be positioned within a slot formed in an external structure (e.g., a cabinet), and methods for installing the same.

### **BACKGROUND**

[0003] Cabinets and other fixtures, furniture, or structures (e.g., wood slat walls) often include integrating lighting (e.g., under cabinet lighting to illuminate a surface or area underneath). Many of these integrated lights are installed in the structure (e.g., cabinet) while is being manufactured or assembled (e.g., before the cabinet has been shipped and installed). In some applications, it is desirable to at least partially construct or assemble the structure in the field (e.g., where it is to be installed) and/or install the lighting device in the field. However, such field constructions have a wider range of tolerances than if the structure were constructed or assembled beforehand (e.g., when forming a slot for receiving the lighting device), and variations in these tolerances can make it difficult to install a lighting device having a fixed width and/or length. Further, field installations of lighting devices often require the installer to additionally install trim or flanges that match the cabinet color and/or finish to cover gaps and/or edges of the lighting device to avoid the housing of the lighting device (e.g., a metal heat sink) from being visible. The present disclosure is directed to solving these and other problems.

### **SUMMARY**

[0004] According to some implementations of the present disclosure, a lighting device includes a housing, a light source, a light guide panel, and a lens. The housing includes a base, a first side wall, a second side wall, a first bumper protruding from an outer surface of the first side wall, and a second bumper protruding from an outer surface of the second side wall. The first bumper and second bumper are configured to aid in positioning the housing within a slot formed in an external structure. The light source is coupled to the housing. The light guide panel is coupled to the housing and configured to diffuse light emitted by the light source. The lens includes a base, a first side wall, a second side wall, a first wing protruding from an inner surface of the first side wall, and a second wing protruding from an inner surface of the second side wall. The lens is coupled to the housing such that a portion of the first side wall of the housing and a portion of the second side wall of the housing are positioned between the first side wall of the lens and the second side wall of the lens.

[0005] According to some implementations of the present disclosure, a lighting device includes a housing, a light source, a light guide panel, and a lens. The housing includes a pair of bumpers configured to aid in positioning the housing in a slot formed in an external structure. The light source is coupled to the housing. The light guide panel is coupled to the housing and configured to diffuse light emitted by the light source. The lens includes a pair of wings configured to aid in distributing light, the lens being coupled to the housing such that a portion of the housing is

disposed within an internal cavity of the lens.

[0006] According to some implementations of the present disclosure, a system includes a plurality of lighting devices. Each of the plurality of lighting devices includes a housing including a pair of bumpers configured to aid in positioning the housing in a slot formed in an external structure, a light source coupled to the housing, a light guide panel coupled to the housing and configured to diffuse light emitted by the light source, and a lens including a pair of wings configured to aid in distributing light, the lens being coupled to the housing such that a portion of the housing is disposed within an internal cavity of the lens.

[0007] According to some implementations of the present disclosure, a method includes positioning a housing of a lighting device within a slot formed in an external structure, the housing including a light source coupled thereto. The method also includes coupling the housing of the lighting device to a mounting surface within the slot via one or more fasteners. The method also includes coupling a light guide panel to the housing subsequent to coupling the housing to the mounting surface. The method also includes coupling a lens to the housing subsequent to coupling the light guide panel to the housing.

[0008] The above summary is not intended to represent each implementation or every aspect of the present disclosure. Additional features and benefits of the present disclosure are apparent from the detailed description and figures set forth below.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1A is an assembled perspective view of a lighting device, according to some implementations of the present disclosure;

[0010] FIG. 1B is an exploded perspective view of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0011] FIG. 2A is a cross-sectional view of a housing of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0012] FIG. 2B is a partial perspective view of the housing of the lighting device, according to some implementations of the present disclosure;

[0013] FIG. 3 is an end view of a lens of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0014] FIG. 4 is a partial perspective view of a light source of the lighting device of FIG. 1, according to some implementations of the present disclosure;

[0015] FIG. 5 is a cross-sectional view of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0016] FIG. 6A is a perspective view of a wiring guide of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0017] FIG. 6B is a perspective view of a wiring guide of the lighting device of FIG. 1A, according to some implementations of the present disclosure;

[0018] FIG. 7 is a process flow diagram for a method for installing one or more lighting devices, according to some implementations of the present disclosure;

[0019] FIG. 8A is a cross-sectional view of an external surface, according to some implementations of the present disclosure;

[0020] FIG. 8B is a cross-sectional view of the external surface and a lighting device including a housing and a light source, according to some implementations of the present disclosure;

[0021] FIG. 8C is a cross-sectional view of the external surface and the lighting device of FIG. 8B further including a light guide panel, according to some implementations of the present disclosure; and

[0022] FIG. 8D is a cross-sectional view of the external surface and the lighting device of FIG. 8C further including a lens, according to some implementations of the present disclosure.

[0023] While the present disclosure is susceptible to various modifications and alternative forms, specific implementations and embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that it is not intended to limit the present disclosure to the particular forms disclosed, but on the contrary, the present disclosure is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure as defined by the appended claims.

#### DETAILED DESCRIPTION

[0024] Referring to FIGS. 1A and 1B, a lighting device **100** according to some implementations of the present disclosure is illustrated. The lighting device **100** includes a housing **110**, a lens **130**, a light source **140**, a light guide panel **150**, a wiring guide **160**, a power connector **170**, and a fastener **180**. As described herein, the lighting device **100** can be mounted with a slot formed in an external structure (e.g., a cabinet or other furniture) such that the lens **130** is generally flush with a finished surface of the external structure.

[0025] Referring to FIGS. 2A and 2B, the housing **110** includes a base **112**, a first side wall **114A**, and a second side wall **114B**. The first side wall **114A** and the second side wall **114B** generally extend from the base **112** and define an internal cavity **116** of the housing **110**. The housing **110** also includes a first bumper **120A** extending from an outer surface of the first side wall **114A** and a second bumper **120B** extending from an outer surface of the second side wall **114B**. The first side wall **114A** also includes a first ridge **118A** on the outer surface thereof that extends along the length of the first side wall **114A**. The second side wall **114B** includes a second ridge **118B** that is the same as, or similar to, the first ridge **118A**. As described herein, the first ridge **118A** and the second ridge **118B** engage a portion of the lens **130** to aid in coupling the lens **130** to the housing **110** (e.g., by a snap fit coupling).

[0026] Generally, the first bumper **120A** and the second bumper **120B** are moveable or deflectable relative to the first side wall **114A** and the second side wall **114B**, respectively, to aid in positioning the housing **110** within an opening during installation of the lighting device **100**. The first bumper **120A** and the second bumper **120B** can extend along at least a portion of the length of the housing **110** (e.g., along the entire length of the housing **110**). The first bumper **120A** includes a first slot **122A** formed therein. The first slot **122A** and the second slot **122B** each have a generally circular or semi-circular cross-section and extend along the length of the first bumper **120A** and the second bumper **120B**, respectively.

[0027] The first slot **122A** and the second slot **122B** can be used for a variety of purposes. For example, in some implementations, the lighting device **100** can be coupled to another lighting device (e.g., that is the same as, or similar to, the lighting device **100**) via alignment pins that are partially received within the first slot **122A** and the second slot **122B** (e.g., forming a press or interference fit). These alignment pins aid in coupling the lighting device **100** to the other lighting device and to align the two lighting devices. In other implementations, the lighting device **100** can include an end cap at either end of the housing **110**. The end cap can comprise the same material as the lens **130** to allow light to pass through. The end cap can be coupled to the housing via one or more fasteners (e.g., screws) that are received within the first slot **122A** and the second slot **122B**.

[0028] The housing **110** also includes a groove **124**, a first slot **126A**, and a second slot **126B** formed in an upper surface of the base **112**. The groove **124**, the first slot **126A**, and/or the second slot **126B** generally extend along the length of the housing **110** (e.g., the entire length of the housing **110**). Like the first slot **122A** in the first bumper **120A** and the second slot **122B** in the second bumper **120B**, the first slot **126A** and the second slot **126B** have a generally circular or semi-circular cross-section and aid in permitting movement or deflection of the first side wall **114A** and/or the second side wall **114B**, respectively, relative to the base **112**. This movement or deflection of the first side wall **114A** and/or second side wall **114B** further aids in positioning the

housing **110** within an opening during installation of the lighting device **100**.

[0029] In some implementations, the groove **124** can receive one or more coupling or attachment mechanisms therein to aid in coupling the lighting device **100** to an external structure (e.g., as shown in FIGS. **8B-8D**) and/or to another lighting device (e.g., that is the same as, or similar to, the lighting device **100**). For example, the lighting device **100** can include an alignment plate or tab that includes a first portion configured to be received within the groove **124** and a second portion configured to be received within the tongue of another lighting device. The alignment plate or tab forms a press or interference fit within the respective tongues to aid in coupling the lighting devices together, and also aids in aligning the two lighting devices. The alignment plate or tab can have any suitable shape to facilitate arranging the two lighting devices, for example, regardless of whether the two lighting devices are arranged with parallel longitudinal axes or at an angle relative to one another (e.g., to form a miter joint or corner). While the lighting device **100** is described herein as being coupled to an external structure using the fastener **180** (e.g., as shown in FIGS. **8B-8D**), in other implementations, the lighting device **100** can be coupled to the external structure using one or more clips (e.g., a spring clip with arms), one more brackets, one or more magnets, or any combination thereof that are received within the groove **124**.

[0030] In some implementations, the housing **110** further includes a cutout **128** (FIG. **2B**) formed in an end of the base **112** within the groove **124**. As described herein, the cutout **128** is sized and shaped to receive a portion of the wiring guide **160** therein to aid in coupling the wiring guide **160** to the housing **110** (e.g., as shown in FIG. **1A**). In some implementations, the housing **110** also includes a scribed line **129** extending along a lower surface of the base **112**. The scribed line **129** is a notch or channel along the center longitudinal axis of the housing **110** that aids in drilling one or more aperture in the base **112** of the housing **110** so that the housing **110** can be mounted to a surface using one or more fasteners.

[0031] The housing **110** comprises a material (e.g., aluminum, steel, stainless steel, etc.) that aids in absorbing and/or distributing heat generated by the light source **140** so that the housing **110** acts a heat sink. As shown, the housing **110** is a unitary and/or monolithic component. In other implementations, the housing **110** can comprise one or more components that are coupled (e.g., welded) together. In some implementations, the housing **110** has a width (measured from the outer edges of the first bumper **120A** and the second bumper **120B**) of about 1.75 inches.

[0032] Referring to FIG. **3**, the lens **130** includes a base **132**, a first side wall **134A**, and a second side wall **134B**. The first side wall **134A** and the second side wall **134B** extend from the base **132** to define an interior cavity **133** of the lens **130**. The first side wall **134A** includes a first hook **136A** and the second side wall **134B** includes a second hook **136B**. As described herein, the first hook **136A** and the second hook **136B** aid in coupling the lens **130** to the housing **110** (e.g., via a snap fit coupling). The lens **130** also includes a first wing **138A** extending from the first side wall **134A** and a second wing **138B** extending from the second side wall **134B**. More specifically, the first wing **138A** extends from an area at or adjacent to the intersection between the first side wall **134A** and the base **132**. The first wing **138A** and the second wing **138B** generally have a curved shaped and extend along the length of the lens **130** (e.g., the entire length of the lens **130**).

[0033] The lens **130** generally comprises a material (e.g., a polymer) for allowing light to pass through so that light emitted from a light source in the internal cavity **133** is emitted outside of the lens **130**. The lens **130** can comprise a generally flexible material that permits deflection of the first side wall **136A** and/or the second side wall **136B** to aid in forming a snap fit coupling between the lens **130** and the housing **110**. In some implementations, the lens **130** includes a frosted surface or material (e.g., a material that is roughened, textured, or patterned) to aid in diffusing light in all directions. In some implementations, the lens **130** has a width of about 1.625 inches and a length of about 100 inches.

[0034] Referring to FIG. **4**, the light source **140** includes a support substrate **142** and a plurality of light emitting diodes (LEDs) **144a-c** coupled to the support substrate **142**. The light source **140** also

includes an encapsulant **146** (e.g., comprising silicon) that at least partially surrounds the plurality of LEDs **144a-c**. The support substrate **142** can be a rigid board or plate, a flexible strip or film, a printed circuit board, etc. The plurality of LEDs **144a-c** can include white LEDs, blue LEDs (e.g., III-nitride LEDs), red LEDs, green LEDs, yellow LEDs, amber LEDs, orange LEDs, infrared LEDs, ultraviolet LEDs, or any combination thereof. The plurality of LEDs **144a-c** can be dimmable (e.g., together or individually controllable). The plurality of LEDs **144a-c** can be mounted to the support substrate **142** using any suitable packaging technique (e.g., a surface mount package, a through-pin package, etc.). Further, the plurality of LEDs **144a-c** can be electrically coupled to one another using any suitable mechanism or technique (e.g., in series, in parallel, via wiring, via traces, etc.). The encapsulant **146** protects the plurality of LEDs **144a-c** (e.g., from mechanical damage) and/or modifies the light emission characteristics. For example, the encapsulant **146** can include a wavelength converter (e.g., a phosphor material) that converts a first wavelength of light emitted by the LEDs **144a-c** (e.g., blue) to a second wavelength of light (e.g., white), which is then emitted from encapsulant **146**.

[0035] Referring to FIG. 5, the light guide panel **150** is coupled to the base **112** of the housing **110** via an adhesive strip **152**. The light guide panel **150** generally aids in diffusing light emitted by the light source **140**. In some implementations, the light guide panel **150** comprises an acrylic material that aids in diffusing light across substantially the entire length and width of the light guide panel **150**. The light guide panel **150** has a generally rectangular profile and extends along substantially the entire length of the housing **110** and/or the light source **140**. The light guide panel **150** does not include any fasteners (e.g., screws) that couple to the housing **110** because such fasteners would cause dark spots where light would not be emitted from the light guide panel. To aid in avoiding such dark spots, the light guide panel **150** is coupled to the housing **110** via the adhesive strip **152**. In some implementations, a first side of the light guide panel **150** includes a laminate or film that further aids in extracting and diffusing light from the light guide panel **150**. In such implementations including the laminate/film, the light guide panel **150** has a width that is about 1.75 inches or less. The film can be a matte polyester film coated with a pressure sensitive acrylic adhesive and a release liner backing. In one non-limiting example, the film on the light guide panel **150** can be a FLEXcon COMPUcall II® PM 100 C MTC-329 L-23 58 PW-8 film.

[0036] While the lighting device **100** is shown as including one adhesive strip **152**, the lighting device **100** can include a plurality of adhesive strips for coupling the light guide panel **150** to the housing **110**. For example, the lighting device **100** can include a plurality of adhesive strips having a length between about 3 inches and about 4 inches positioned about every 12 inches along the length of the light guide panel **150**. In some implementations, the light guide panel **150** includes a first edge that is clear and a second opposing edge that includes reflective (e.g., silver or white) tape. The reflective tape prevents light from escaping the second edge of the light guide panel and reflects the light back towards the first edge, thereby increasing light diffusion and extraction out of the light guide panel **150**.

[0037] The adhesive strip **152** is a double-sided adhesive that adheres to both the housing **110** and the light guide panel **150**. In some implementations, the adhesive strip **152** can have a width of about 0.75 inches and a length of about 3 inches. In some cases, heat generated by the light guide panel **150** may cause degradation of the adhesive strip **152** (e.g., reducing its holding power). Thus, the adhesive strip **152** preferably has a heat resistance rating of about 120 degrees Fahrenheit. To secure the light guide panel **150** to the housing **110**, the adhesive strip **152** can have a peel adhesion that is between about 35 oz/in (N/100 mm) and about 70 oz/in (N/100 mm). The adhesive strip **152** can have a first side having a first peel adhesion value and a second opposing side having a second peel adhesion value that is different than the first peel adhesion value to aid in securing the light guide panel **150** to the housing **110**. While the lighting device **100** is shown as including one adhesive strip **152**, more generally, the lighting device **100** can include a plurality of adhesive strips that are the same as, or similar to, the adhesive strip **152**.

[0038] In some implementations, the adhesive strip **152** is a 3M™ Flexographic Plate Mounting Tape or a 3M™ Cushion-Mount™ Printing Tape, which are generally used for mounting flexographic print plates to cylinders or sleeves. Such double-sided tapes are particularly advantageous for coupling the light guide panel **150** to the housing **110**, for example, due to their holding power and heat resistance. As one non-limiting example, the adhesive strip **152** can be a 3M™ Cushion-Mount™ Plus Standard Combination Plate Mounting Tape, Product Number E1040H.

[0039] The light source **140** is coupled to an inner surface of the second side wall **114B** of the housing **110** (e.g., using an adhesive strip that is the same as, or similar to, the adhesive strip **152**). While the light source **140** generally emits light in a plurality of directions, a majority of the light is generally emitted towards the first side wall **114A** opposite the second side wall **114B**. This light enters the light guide panel **150**, which generally distributes (e.g., diffuses) the light in a direction that is generally opposite the base **112** of the housing **110** and towards the lens **130**. As shown in FIG. 5, the first wing **138A** of the lens and the second wing **138B** of the lens **130** generally extend from the base **132** of the lens **130** to the light guide panel **150**. In other words, the first wing **138A** generally extends from the intersection between the first side wall **134A** and the base **132** of the lens **130** and the light guide panel **150**. Similarly, the second wing **138B** generally extends from the intersection between the second side wall **134B** and the base **132** of the lens **130** and the light guide panel **150**. The first wing **138A** and the second wing **138B** aid in distributing or diffusing light, thereby aiding in preventing or reducing dark spots at the edges or corners of the lens **130**.

[0040] In some examples, light emitted from the light source **140** may spill out onto the lens **130** without entering the light guide panel **150** if the light source **140** is not sufficiently aligned with the light guide panel **150**. However, even in this case, the first wing **138A** and the second wing **138B** also aid in diffusing light that spills out from the light source **140** (e.g., aiding in preventing or reducing light hot spots on the lens **130**), thereby reducing the required tolerance for the relative positions of the light source **140** and the light guide panel **150**.

[0041] Referring still to FIG. 5, the lens **130** is coupled to the housing **110** via a snap fit coupling. As described herein, this snap fit coupling allows an installer to couple the lens **130** to the housing **110** after the housing **110** is mounted to an external structure (e.g., a cabinet). To form the snap fit coupling, the first hook **136A** of the first side wall **134A** of the lens **130** engages the first ridge **118A** of the first side wall **114A** of the housing **110**. Similarly, the second hook **136B** of the second side wall **134B** of the lens **130** engages the second ridge **118B** of the second side wall **114B** of the housing **110**. As shown in FIG. 5, a portion of the first side wall **114A** and a portion of the second side wall **114B** are disposed within the internal cavity **133** of the lens **130**. In other words, a portion of the lens **130** surrounds a portion of the housing **110** so that the portion of the housing **110** is not visible from outside of the lens **130**.

[0042] While the lighting device **100** is shown as including one light source **140**, in some implementations, the lighting device **100** includes a plurality of light sources that are the same as, or similar to, the light source **140**. In such implementations, a first one of the light sources can be coupled to the inner surface of the first side wall **114A** and a second one of the light sources can be coupled to the inner surface of the second side wall **114B**. The first light source can emit a first wavelength of light (e.g., white light) and the second light source can emit a second wavelength of light or a plurality of wavelengths of light (e.g., red light, blue light, green light, etc.). The plurality of light sources can be tuned or adjusted to create a white or warm dim effect that avoids the pink effect that is visible when the color delta is greater than 2000K and the dimming curve no longer follows the black body dimming curve.

[0043] Referring to FIGS. 6A and 6B, the wiring guide **160** is generally used to guide the power cable **172** through the housing **110** so that the connector **170** can be coupled to (e.g., directly or indirectly) a power supply (e.g., an AC electrical outlet). The wiring guide **160** includes an aperture **162**, a slot **164**, a protrusion **166**, and a tongue **168**. The aperture **162** and slot **164** are sized and

shaped to allow at least a portion of the power cable **172** to pass through. The protrusion **166** extends from a lower surface of the wiring guide **160** so that a portion of the power cable **172** can be wrapped around the protrusion **166** to aid in keeping the power cable **172** in place. The tongue **168** is sized and shaped to engage the groove **124** of the housing **110** (FIGS. 2A-2B) so that the wiring guide **160** can be coupled to the cutout **128** (FIG. 2B) of the housing **110**, as shown in FIG. 1A. As discussed herein, in some implementations, the lighting device **100** includes a plurality of light sources (e.g., one coupled to the first side wall **120A** and a second coupled to the second side wall **120B**). In such implementations, the wiring guide **160** can include one or more additional apertures so that the wiring for both light sources can be received within and fed through the wiring guide **160**. While the wiring guide **160** is shown as including one tongue **168**, in other implementations, the wiring guide **160** can additionally include a second tongue that is received within a groove (e.g., like the groove **124**) of another lighting device that is the same as, or similar to, the lighting device **100**. This aids in coupling the lighting device and the other lighting device as well as aligning the two lighting devices.

[0044] Referring back to FIGS. 1A and 1B, the ends of the lighting device **100** are generally straight (e.g., the housing **110** and the lens **130** have a generally rectangular profile). However, in some implementations, the lighting device **100** can include one or more mitered ends. In such implementations, the mitered end(s) appear as if cut at an angle that is, for example, between about 30 degrees and about 80 degrees (e.g., 45 degrees, 60 degrees, 75 degrees, etc.). A first lighting device with a mitered end can be positioned to abut another lighting device with a mitered end to form a miter joint such that a longitudinal axis of the first lighting device is at angle (e.g., 90 degrees) relative to a longitudinal axis of the second lighting device. In such implementations, the light source **140** is positioned on the longer edge of the housing **110**. The light guide panel **150** is additionally advantageous in such implementations to prevent or reduce the appearance of a dark line or edge at the miter joint between the two lighting devices.

[0045] Referring to FIG. 7, a method **700** for installing a light device according to some implementations of the present disclosure is illustrated. The method **700** can be performed using, for example, the lighting device **100** described herein.

[0046] Step **701** of the method **700** includes positioning a housing of a lighting device within a slot formed in an external structure. The external structure can be, for example, a cabinet or other furniture. Referring to FIG. 8A, an external structure **800** includes millwork portion **802**, millwork portion **804**, blocking false bottom portion **806**, and blocking false bottom portion **808**. The external structure **800** also includes a slot **820** formed between the millwork portion **802** and millwork portion **804** and an opening **822** between the blocking false bottom portion **806** and the blocking false bottom portion **808**. The slot **820** can be a generally rectangular slot formed in the external structure having a width of, for example, about 1.75 inches. The blocking false bottom portion **806** includes a mounting surface **810** to which the lighting device **100** is mounted.

[0047] The lighting device **100** can be positioned within the slot **820** as shown in FIG. 8B. As described herein, the first bumper **120A** and/or the second bumper **120B** are moveable or deflectable relative to the base **112** of the housing **110** to aid in positioning the housing **110** within the slot **120**. Similarly, in some implementations, the first side wall **114A** and/or the second side wall **114B** are moveable or deflectable relative to the base **112** to further aid in positioning the housing **110** within the slot **120**. If the housing **110** has a width of about 1.75 inches and the slot **820** has a width of about 1.75, this movement or deflection of portions of the housing **110** allow the housing **110** to be received and secured within the slot **820** even if the spacing or tolerance for the slot **820** is inaccurate. This is particularly advantageous in field constructions where the slot **820** is formed in the field (e.g., not in the factory where the external structure **800** is made) and where the lighting device **100** is attached in the field.

[0048] In some implementations, step **701** includes marking a location for the opening **822** based on the position of the power cable **172** when the housing **110** is positioned within the slot **820** and

forming (e.g., drilling) the opening **822** between the blocking false bottom portion **806** and the blocking false bottom portion **808**. While the opening **822** is shown to the right of the fastener **180** in FIG. **8B**, more generally, the opening **822** can be formed at any suitable location relative to the housing **110** and fastener **180** (e.g., to the left of the fastener **180** shown in FIG. **8B**).

[0049] Step **702** of the method **700** includes coupling the housing of the lighting device to a mounting surface within the slot via one or more fasteners. For example, referring to FIG. **8B**, the housing **110** is coupled to the mounting surface **810** via the fastener **180** (e.g., a threaded screw), which extends through the housing **110** and engages the blocking false bottom portion **808** to aid in securing the housing **110** within the slot **820**. As shown, during step **702**, the light source **140** is coupled to the housing **110**, but the lens **130** and the light guide panel **150** are not. In some implementations, step **702** includes drilling one or more apertures in the base **112** of the housing **110** and/or through the mounting surface **810** for receiving the one or more fasteners therein. For example, step **702** can include drilling a plurality of apertures along the scribed line **129** (FIG. **2A**) of the base **112** of the housing **110** and corresponding apertures through the mounting surface **810**. In this example, the drilling of the apertures can be performed while the housing **110** is positioned within the slot **820** generally as shown in FIG. **8B**.

[0050] Step **703** of the method **700** includes coupling a light guide panel to the housing subsequent to coupling the housing to the mounting surface. For example, step **702** can include coupling one or more adhesive strips to the housing. The adhesive strips can be reusable (e.g., attached to one location on the housing, removed, then attached to another location on the housing). Referring to FIG. **8B**, the adhesive strip **152** is adhered to the housing **110**. Then, the light guide panel **150** is adhered to the housing **110** via the adhesive strip **152**. In some implementations, the light guide panel **150** includes a clear edge and a silver tape edge. In such implementations, step **703** includes pushing the clear edge of the light guide panel **150** against the light source **140** at an angle, centering the light guide panel **150** along light source **140**, and then rotating the light guide panel **150** relative to the housing **110** and pushing towards the adhesive strip **152**.

[0051] Step **704** of the method **700** includes coupling a lens to the housing subsequent to coupling the light guide panel to the housing. For example, referring to FIG. **8D**, the lens **130** is coupled to the housing **110**. As described herein, the first hook **136A** of the first side wall **134A** of the lens **130** engages the first ledge **118A** of the first side wall **114A** of the housing **110** and the second hook **136B** of the second side wall **134B** of the lens **130** engages the second ledge **118B** of the second side wall **114B** of the housing **110** to form a snap fit coupling. In some implementations, step **704** includes cutting the lens **130** so that the length of the lens **130** is substantially equal to the length of the housing **110**. In some cases, the millwork portion **802** and/or millwork portion **804** may be moved after the housing **110** has been installed; however, the first bumper **122A** and second bumper **122B** prevent the housing **110** from being compressed and ensure sufficient spacing between the edges of the millwork portion **802** and/or millwork portion **804** so that the lens **130** can be snap fit onto the housing **110**.

[0052] As shown in FIG. **8D**, the surface of the lens **130** is generally flush with the bottom surface of the millwork portion **802** and millwork portion **804**. In some implementations, the first side wall **134A** of the lens **130** can be spaced from the millwork portion **802** and the second side wall **134B** of the lens **130** can be spaced from the millwork portion **804** by about 0.0625 inches.

Advantageously, the housing **110** will not be visible when the lens **130** is installed because the first side wall **134A** and the second side wall **134B** of the lens **130** surround at least a portion of the first side wall **114A** and second side wall **114B** of the housing **110**. This eliminates the need to include flanges or an edge of the lighting device that have the same color and/or finish (e.g., nature, white, black, etc.) as the millwork portions **802** and **804**, providing a uniform appearance without mismatched trim.

[0053] One or more elements or aspects or steps, or any portion(s) thereof, from one or more of any of the claims below can be combined with one or more elements or aspects or steps, or any

portion(s) thereof, from one or more of any of the other claims or combinations thereof, to form one or more additional implementations and/or claims of the present disclosure.

[0054] While the present disclosure has been described with reference to one or more particular embodiments or implementations, those skilled in the art will recognize that many changes may be made thereto without departing from the spirit and scope of the present disclosure. Each of these implementations and obvious variations thereof is contemplated as falling within the spirit and scope of the present disclosure. It is also contemplated that additional implementations according to aspects of the present disclosure may combine any number of features from any of the implementations described herein.

## Claims

**1-28.** (canceled)

**29.** A method comprising: positioning a housing of a lighting device within a slot formed in an external structure, the housing including a light source coupled thereto; coupling the housing of the lighting device to a mounting surface within the slot via one or more fasteners; coupling a light guide panel to the housing subsequent to coupling the housing to the mounting surface; and coupling a lens to the housing subsequent to coupling the light guide panel to the housing.

**30.** The method of claim 29, wherein the coupling the lens to the housing includes snap fitting the lens to the housing.

**31.** The method of claim 30, wherein the coupling the light guide panel to the housing includes snap fitting a first side wall of the lens to a first side wall of the housing and snap fitting a second side wall of the lens to a second side wall of the housing.

**32.** The method of claim 29, wherein the external structure includes millwork and a false bottom, wherein the slot is formed in the millwork.

**33.** The method of claim 32, wherein the lens is coupled to the housing such that the lens is generally flush with a finished surface of the millwork.

**34.** The method of claim 33, wherein the false bottom includes an aperture for permitting a power cable of the lighting device to pass through.

**35.** The method of claim 29, wherein the lens is coupled to the housing such that a first side wall of the lens is spaced from a first edge of the slot by a predetermined distance and a second side wall of the lens is spaced from a second edge of the slot by the predetermined distance.

**36.** The method of claim 35, wherein the predetermined distance is about 0.0625 inches.

**37.** The method of claim 29, wherein the coupling the light guide panel to the housing includes: positioning an edge of a light guide panel against the light source; and moving the light guide panel to cause the light guide panel to be adhered to the housing.

**38.** The method of claim 37, wherein the moving the light guide panel includes rotating the light guide panel relative to the housing.

**39.** The method of claim 29, further comprising: prior to coupling the housing to the mounting surface, forming an aperture in the mounting surface; and feeding a feed wire of the lighting device through the aperture in the mounting surface.

**40.** The method of claim 39, further comprising electrically coupling the feed wire to a power supply.

**41.** The method of claim 29, further comprising prior to coupling the housing to the mounting surface, forming one or more apertures in the mounting surface, each of the one or more apertures being configured to receive a corresponding one of the one or more fasteners therein.

**42.** The method of claim 29, wherein the light source includes a substrate, one or more of light emitting diodes (“LEDs”) coupled to the substrate, and an encapsulant at least partially surrounding the one or more LEDs.

**43.** A method comprising: positioning a housing of a lighting device within a slot formed in an

external structure, the housing including a light source; coupling a light guide panel to the housing subsequent to positioning the housing within the slot formed in the external structure; and coupling a lens to the housing subsequent to coupling the light guide panel to the housing, wherein (i) the lens includes including a base, a first side wall, a second side wall, a first wing protruding from an inner surface of the first side wall, and a second wing protruding from an inner surface of the second side wall and (ii) subsequent to coupling the lens to the housing, a portion of the first side wall of the housing and a portion of the second side wall of the housing are positioned between the first side wall of the lens and the second side wall of the lens.

**44.** The lighting device of claim 43, wherein the first wing of the lens extends between the first side wall of the lens and a surface of the light guide panel and the second wing of the lens extends between the second side wall of the lens and the surface of the light guide panel, and wherein the first wing and the second wing of the lens are configured to aid in distributing light emitted by the light source

**45.** The lighting device of claim 42, wherein the first wing has a first radius of curvature and the second wing has a second radius of curvature that is the same as the first radius of curvature.

**46.** The method of claim 43, further comprising coupling the housing of the lighting device to a mounting surface within the slot, wherein the coupling the housing to the mounting surface includes: forming a plurality of apertures in the mounting surface corresponding to a plurality of apertures formed in the housing of the lighting device; and positioning a plurality of fasteners through the plurality of apertures formed in the housing of the lighting device and the plurality of apertures in the mounting surface to couple the housing of the lighting device to the mounting surface.

**47.** The method of claim 46, wherein the coupling the light guide panel to the housing is subsequent to the coupling the housing of the lighting device to the mounting surface and includes: positioning a clear edge of the light guide panel against the light source at an angle; centering the light guide panel relative to the light source; and moving the light guide panel relative to the housing such that the light guide panel contacts an adhesive strip coupled to the housing that is configured to aid in coupling the light guide panel to the housing.

**48.** The method of claim 43, wherein the housing of the lighting device includes a base, a first side wall, and a second side wall, wherein (i) the light source is coupled to the first side wall or the second side wall of the housing and (ii) subsequent to the coupling the light guide panel to the housing, the light guide panel is coupled to the base of the housing.

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